

# Matrix Inverse

**Geometric Algorithms**  
**Lecture 10**

# Practice Problem

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_1 - x_2 \\ 4x_2 \\ 3x_1 - 3x_2 \end{bmatrix}$$

*Determine if the above transformation is onto, one-to-one, both, or neither*

# Answer

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 2x_1 - x_2 \\ 4x_2 \\ 3x_1 - 3x_2 \end{bmatrix}$$



# Objectives

- » Define other several important matrix operations
- » Motivate and define matrix inverses

# Keywords

Matrix Transpose

Inner Product

Matrix Power

Square Matrix

Matrix Inverse

Invertible Transformation

1-1 Correspondence

`numpy.linalg.inv`

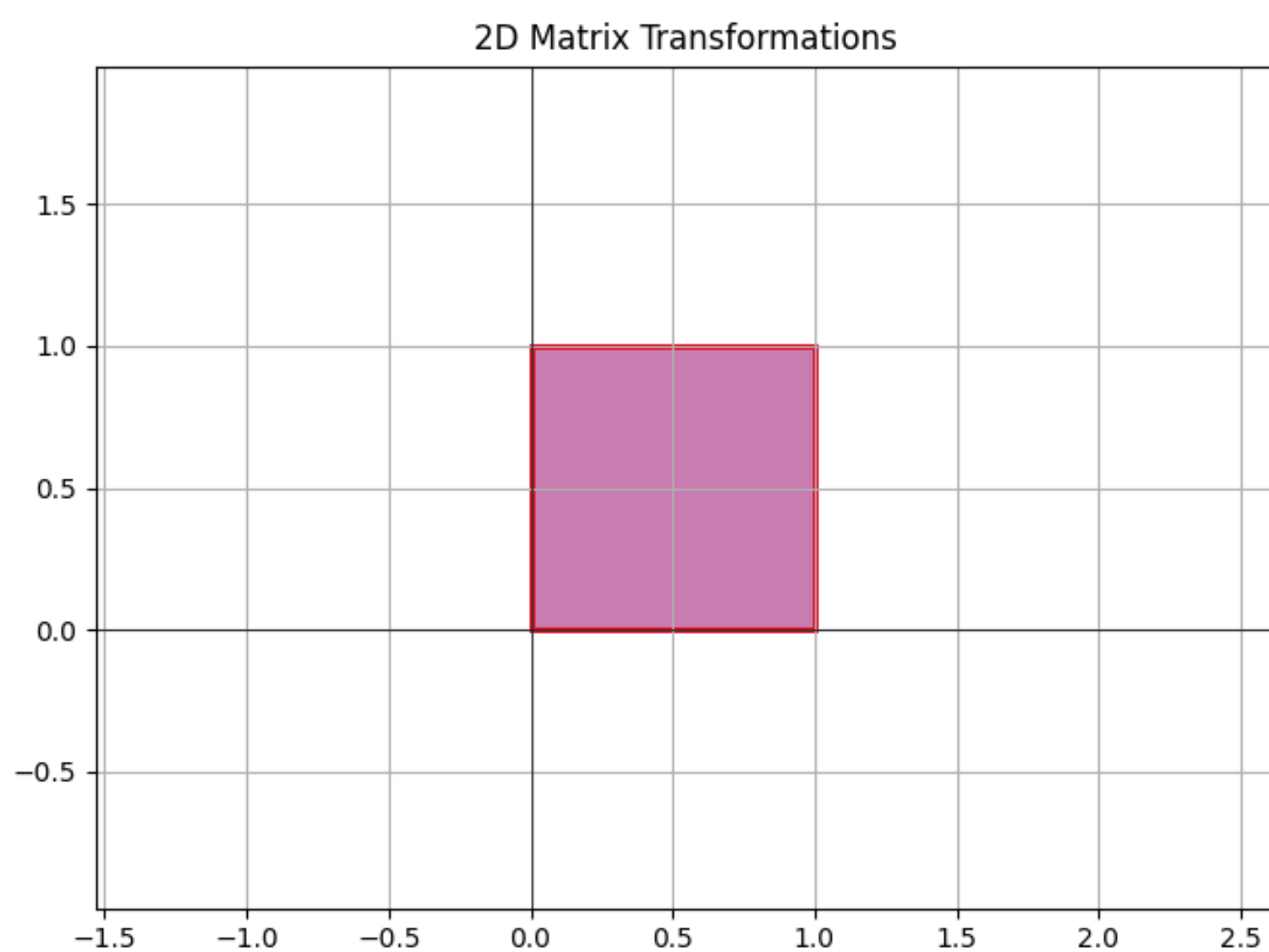
Determinant

Invertible Matrix Theorem

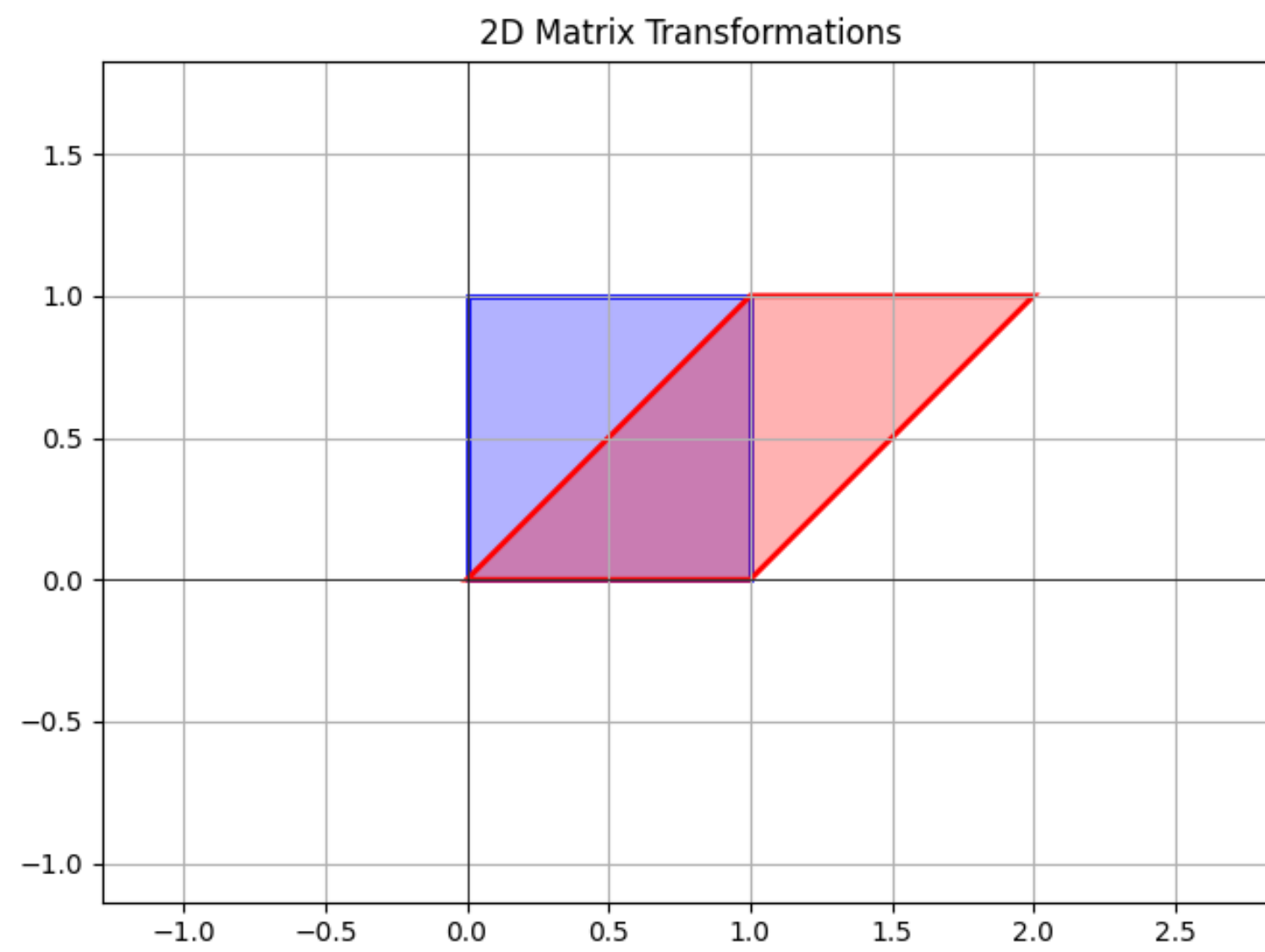
# Recap



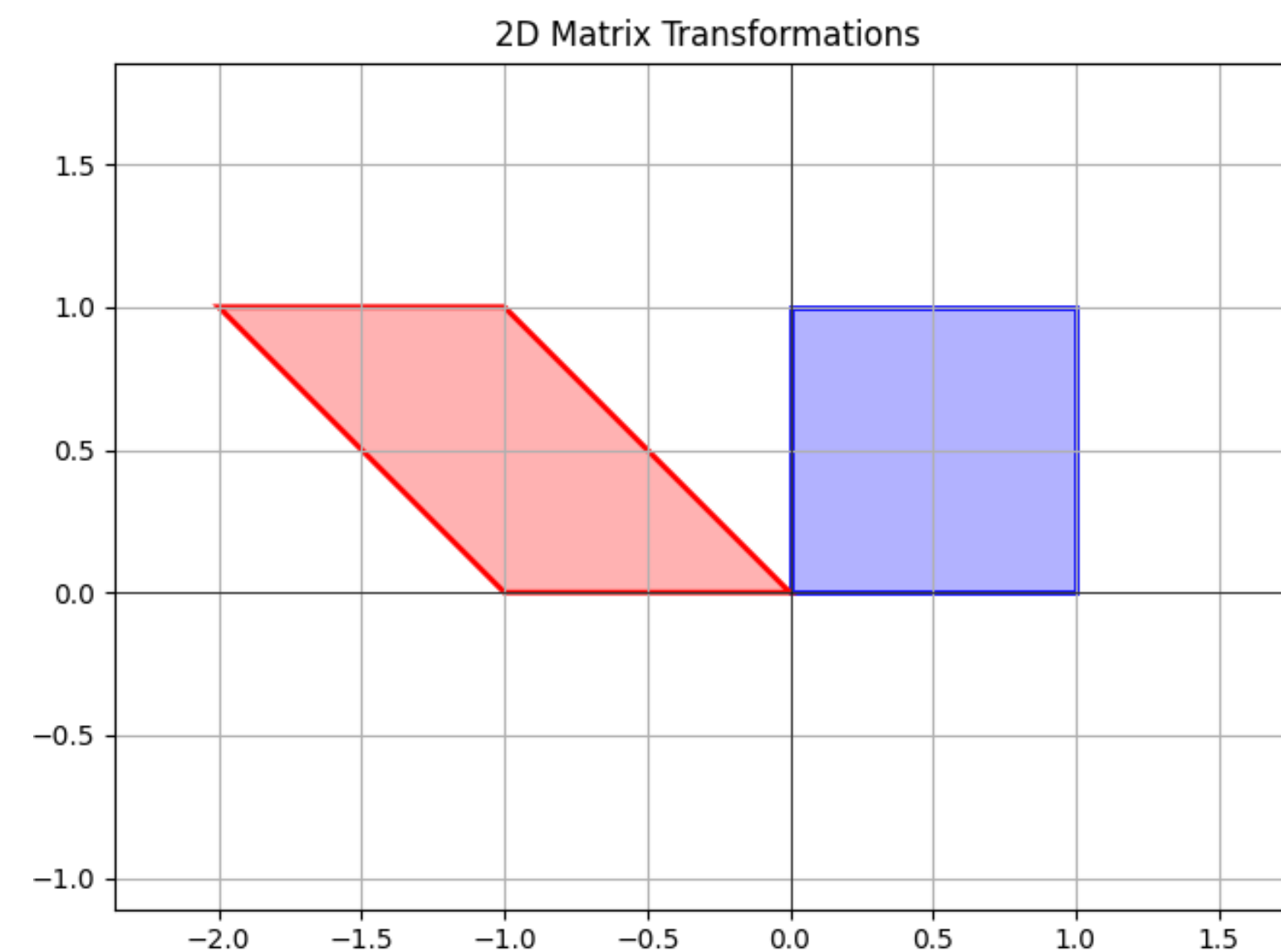
# Recap: Shearing and Reflecting (Geometrically)



shear



reflect



# Recap: The Key Fact

**Thm.** The composition of two linear transformations is a linear transformation

**This means the composition of two matrix transformations can be implemented by a *single* matrix**



# Matrix Multiplication

**Def.** For  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{n \times p}$  with columns  $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$  the product  $AB \in \mathbb{R}^{m \times p}$  is given by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix}$$

Replace each column of  $B$  with  $A$  multiplied by that column

# Recap: Matrix Multiplication and Composition

**Thm.** For any matrices  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{n \times p}$  and any vector  $\mathbf{v} \in \mathbb{R}^p$ ,

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

The matrix implementing the composition is the product of the two underlying matrices

# Recap: Matrix "Interface"

multiplication

what does  $AB$  mean when  $A$  and  $B$  are matrices?

addition

what does  $A + B$  mean when  $A$  and  $B$  are matrices?

scaling

what does  $cA$  mean when  $A$  is matrix and  $c$  is a real number?



# More Matrix Operations

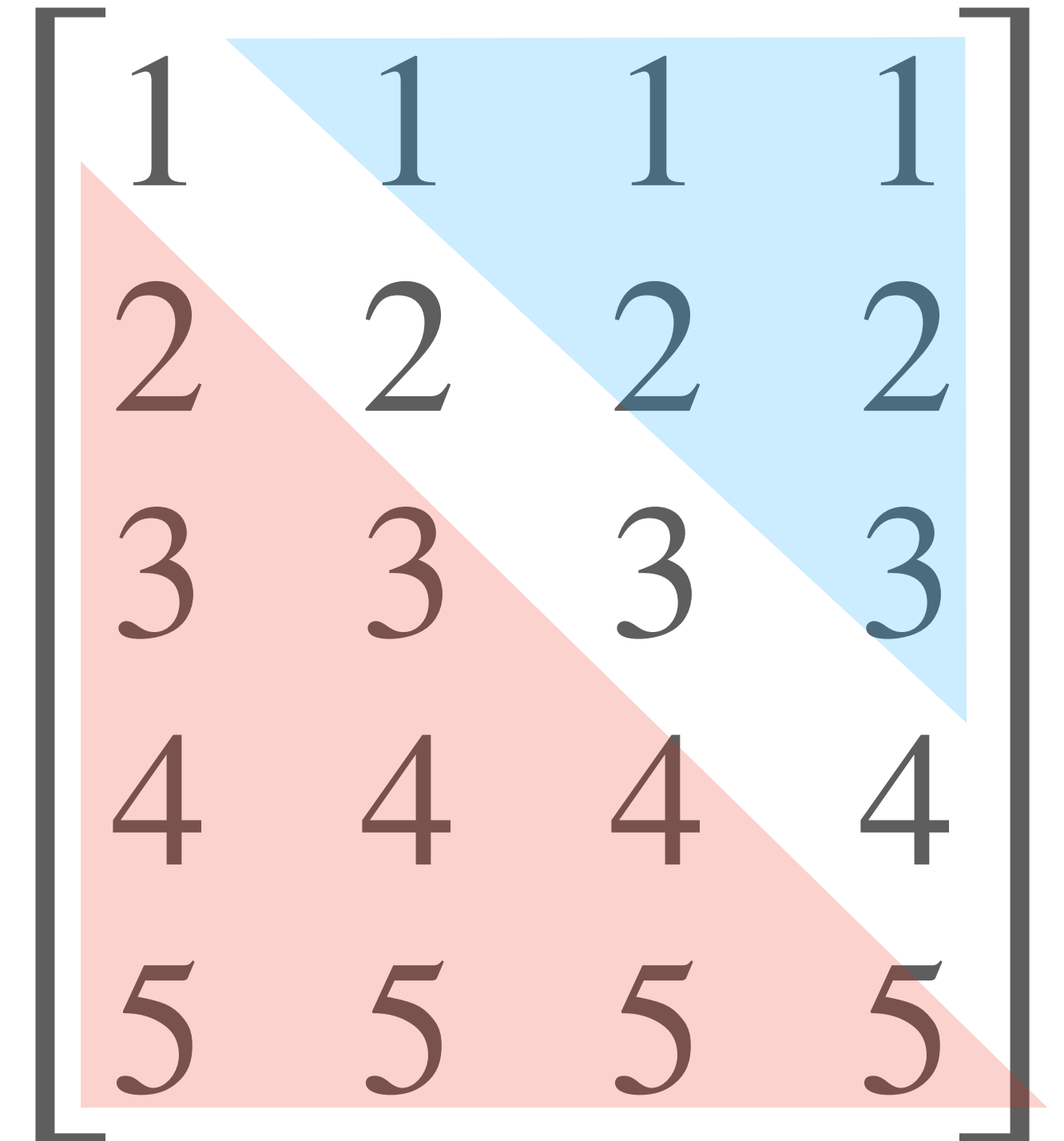
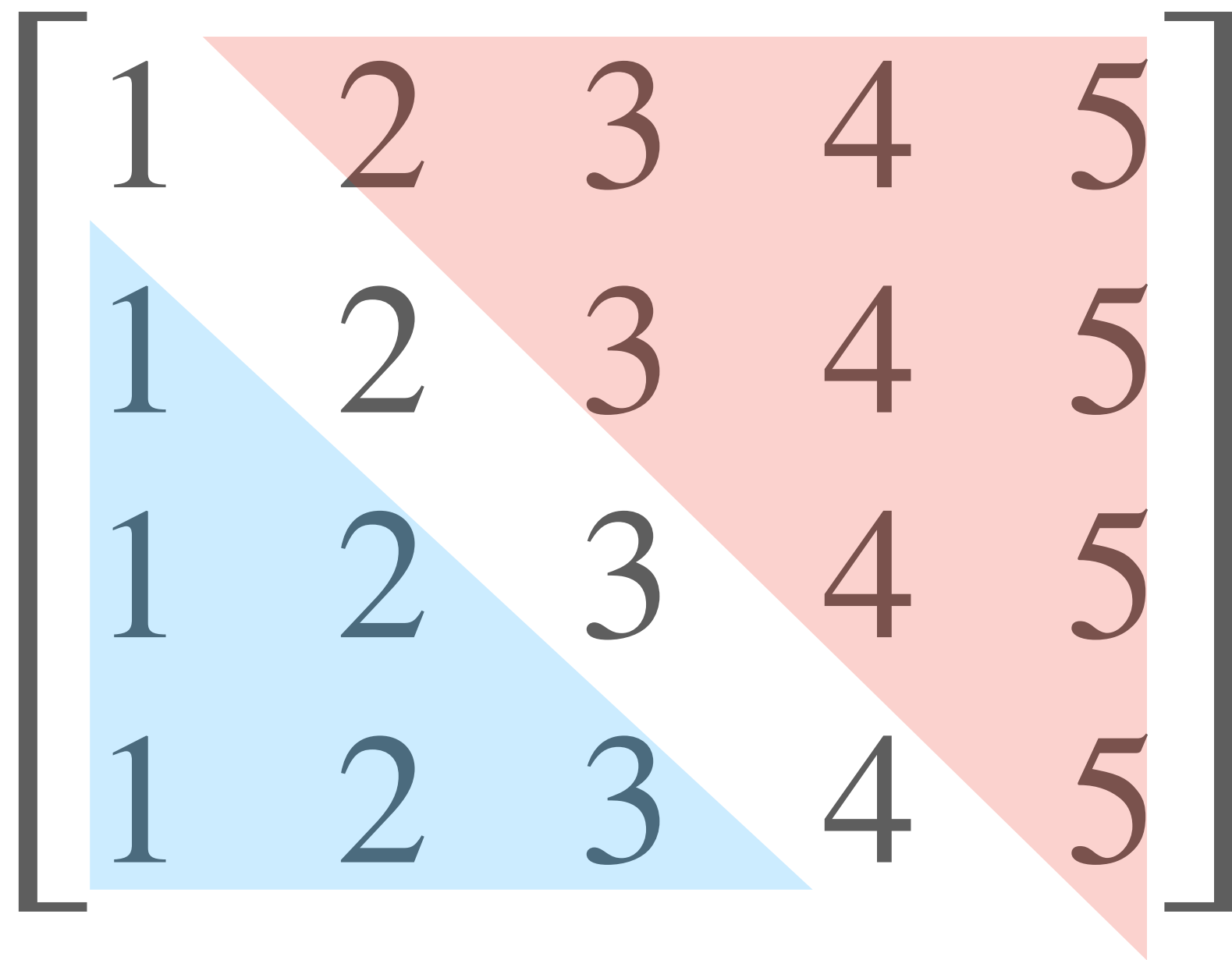
# Transpose (Pictorially)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \end{bmatrix}$$

# Transpose (Pictorially)





# Transpose

**Definition.** For a  $m \times n$  matrix  $A$ , the **transpose** of  $A$ , written  $A^T$ , is the  $n \times m$  matrix such that

$$(A^T)_{ij} = A_{ji}$$

**Example.**

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

# Algebraic Properties (Transpose)

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T$$



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$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T \text{ Important: the order reverses!}$$

# Challenge Problem

*Prove that  $(AB)^T = B^T A^T$  for  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{n \times p}$*

# Transposes and Inner Products

For  $\mathbf{v} \in \mathbb{R}^n$  note that  $\mathbf{v}^T \in \mathbb{R}^{1 \times n}$  is a *single-row matrix*, sometimes called a **row vector**

*For two vectors  $\mathbf{u}$  and  $\mathbf{v}$  in  $\mathbb{R}^n$ , is  $\mathbf{u}^T \mathbf{v}$  defined?*



# Transposes and Inner Products

$$[u_1 \quad u_2 \quad u_3 \quad u_4] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

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**Def.** The **inner product** of two vectors **u** and **v** in  $\mathbb{R}^n$  is  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$

# Matrix Powers



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*What should  $A^0$  be? (we want  $A^0 A^k = A^{0+k} = A^k$ )*

# Matrix Powers (Computationally)

Use `numpy.linalg.matrix_power`

This can be *much* faster than doing a sequence of matrix multiplications (*why?*)

# **Final Warnings about Matrix Multiplication**



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2. If  $AB = AC$  then it is not necessary that  $B = C$
3. If  $AB = 0$  (the zero matrix) it is not necessarily the case that  $A = 0$  or  $B = 0$



# Matrix Operations (So Far)

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transpose

$$A^T$$



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$$cA$$

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addition (subtraction)

$$A + B$$

$$A + (-1)B = A - B$$

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$$A^T$$

scaling

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multiplication (powers)

$$AB$$

$$A^k$$



# Matrix Inverses

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It is the "1" of matrices. For any  $A$

$$IA = AI = A$$

These may be different sizes



# Recall: The Identity Matrix

$$\begin{array}{ccccc} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} & = & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} & \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & = & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \\ 2 \times 2 & 2 \times 4 & & 2 \times 4 & 4 \times 4 & & 2 \times 4 \end{array}$$

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**Definition.** The  $n \times n$  **identity matrix** is the matrix whose *diagonal* contains all 1s, and all other entries are 0s

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$



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**Example.**

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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$\frac{1}{2}$  is the **reciprocal** or **multiplicative inverse** of 2

# Basic Algebra

$$2^{-1}(2x) = 2^{-1}(10)$$

How do we solve this equation?

Divide on both sides by 2 to get  $x = 5$

*Multiply each side by  $\frac{1}{2}$  a.k.a.  $2^{-1}$*

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# Basic Algebra

$$1x = 5$$

How do we solve this equation?

Divide on both sides by 2 to get  $x = 5$

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# Basic Algebra

$$x = 5$$

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Divide on both sides by 2 to get  $x = 5$

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**Wouldn't it be nice...**

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$$

How do we solve this equation?

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Do all matrices have  
inverses?

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inverses?

**No.** If they did, then every linear  
system would have a solution

When does a matrix have  
an inverse?



# Square Matrices

**Def.** A matrix  $A \in \mathbb{R}^{m \times n}$  is **square** if  $m = n$

$$\begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

i.e., it has same number of rows as columns

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» *that can have inverses*

# Matrix Inverses



# Matrix Inverses

**Def.** For a matrix  $A \in \mathbb{R}^{n \times n}$ , an **inverse** of  $A$  is a matrix  $B \in \mathbb{R}^{n \times n}$  such that

$$AB = I_n \text{ and } BA = I_n$$

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$A$  is **invertible** if it has an inverse. Otherwise it is **singular**



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**Example.**  $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

# Example

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

# Example

*Reflection across the  $x$ -axis is it's own inverse*



# Non-Example

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

# Inverses are Unique

**Thm.** If  $B$  and  $C$  are inverses of  $A$ , then  $B = C$

If  $A$  is invertible, then we write  $A^{-1}$  for *the* inverse of  $A$

# Solutions for Invertible Matrix Equations

**Thm.** For matrix  $A \in \mathbb{R}^{n \times n}$ , if  $A$  is invertible then the equation  $A\mathbf{x} = \mathbf{b}$  has a unique solution for any choice of  $\mathbf{b}$



# Unique Solutions

If  $A\mathbf{x} = \mathbf{b}$  has a unique solution for any choice of  $\mathbf{b}$ , then it has

» exactly one solution for any choice of  $\mathbf{b}$

# Unique Solutions

If  $A\mathbf{x} = \mathbf{b}$  has a unique solution for any choice of  $\mathbf{b}$ , then it has

» at least one solution for any choice of  $\mathbf{b}$

» at most one solution for any choice of  $\mathbf{b}$



# Unique Solutions

If  $A\mathbf{x} = \mathbf{b}$  has a unique solution for any choice of  $\mathbf{b}$ , then it has

»  $T$  is onto

»  $T$  is one-to-one

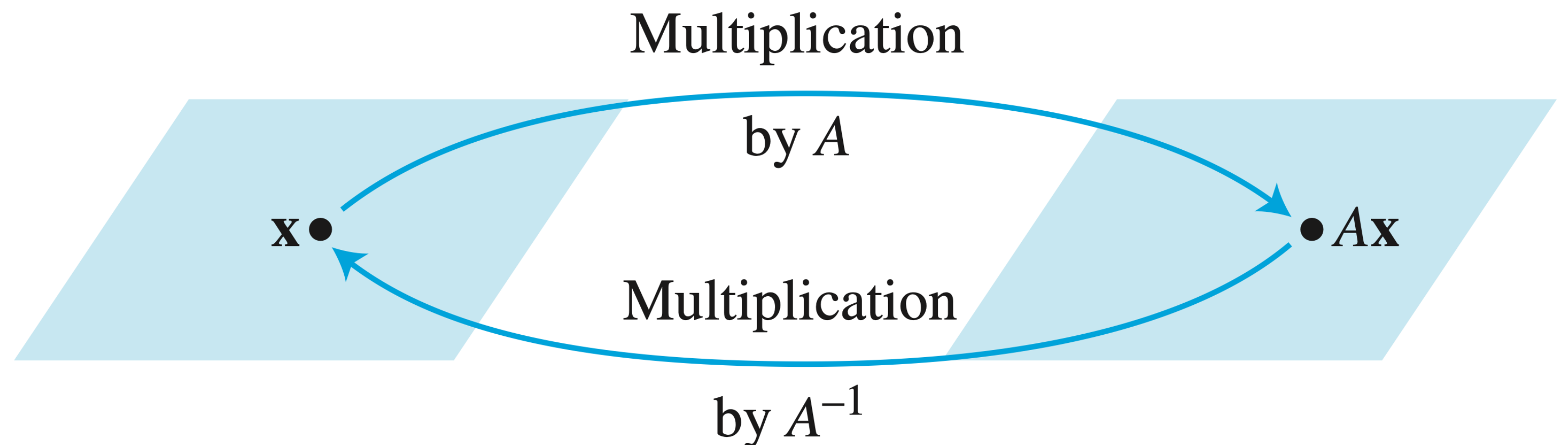
where  $T$  is implemented by  $A$

# Connection to Transformations

**Def.** A linear transformation  $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$  is **invertible** if there is a linear transformation  $S$  such that

$$S(T(\mathbf{v})) = \mathbf{v} \quad \text{and} \quad T(S(\mathbf{v})) = \mathbf{v}$$

for any  $\mathbf{v}$  in  $\mathbb{R}^n$



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**Thm.** A matrix  $A \in \mathbb{R}^{n \times n}$  is invertible if and only if the matrix transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is invertible

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**Non-Example.** Projection onto the  $x_1$ -axis in  $\mathbb{R}^2$



# Non-Example

*Projection onto the  $x$ -axis in  $\mathbb{R}^2$*

# Connection to Transformations



# Connection to Transformations

**Def.** A transformation  $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$  is a **one-to-one correspondence** (bijection) if any vector  $\mathbf{b}$  in  $\mathbb{R}^n$  is the **image of exactly one vector**  $\mathbf{v}$  in  $\mathbb{R}^n$  (where  $T(\mathbf{v}) = \mathbf{b}$ )

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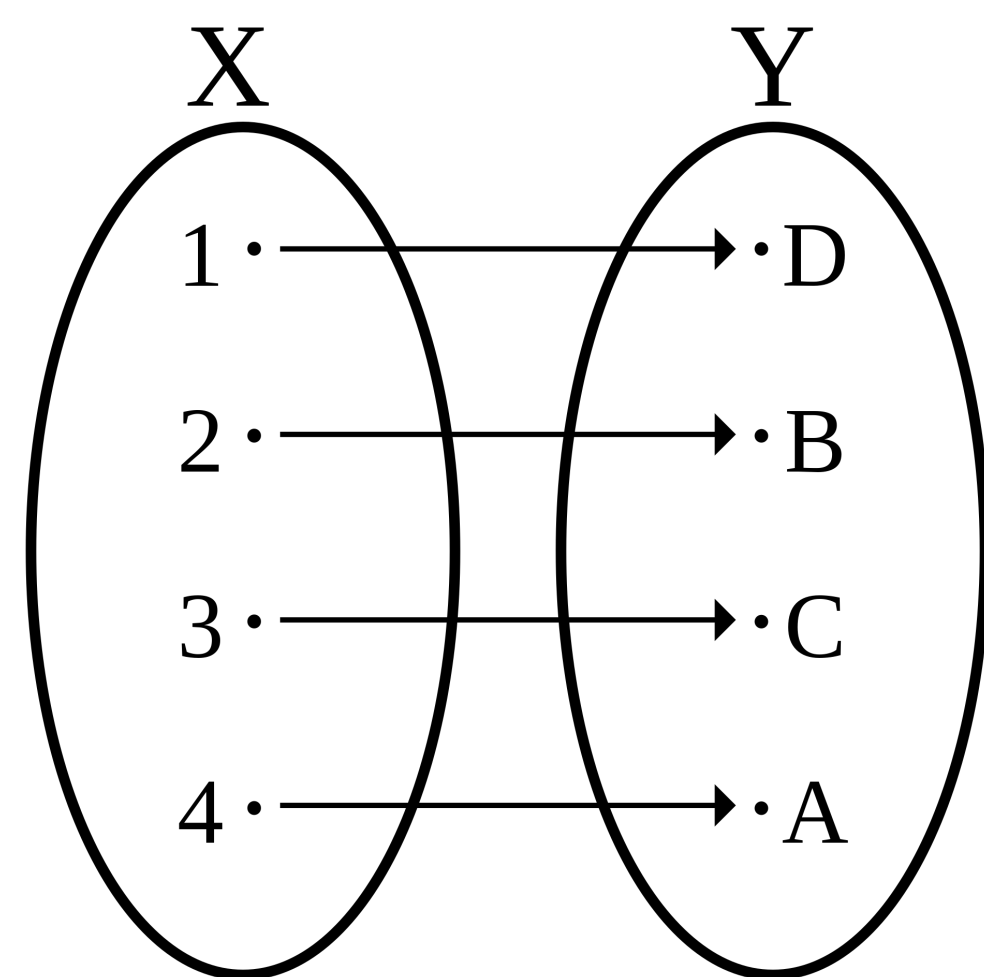
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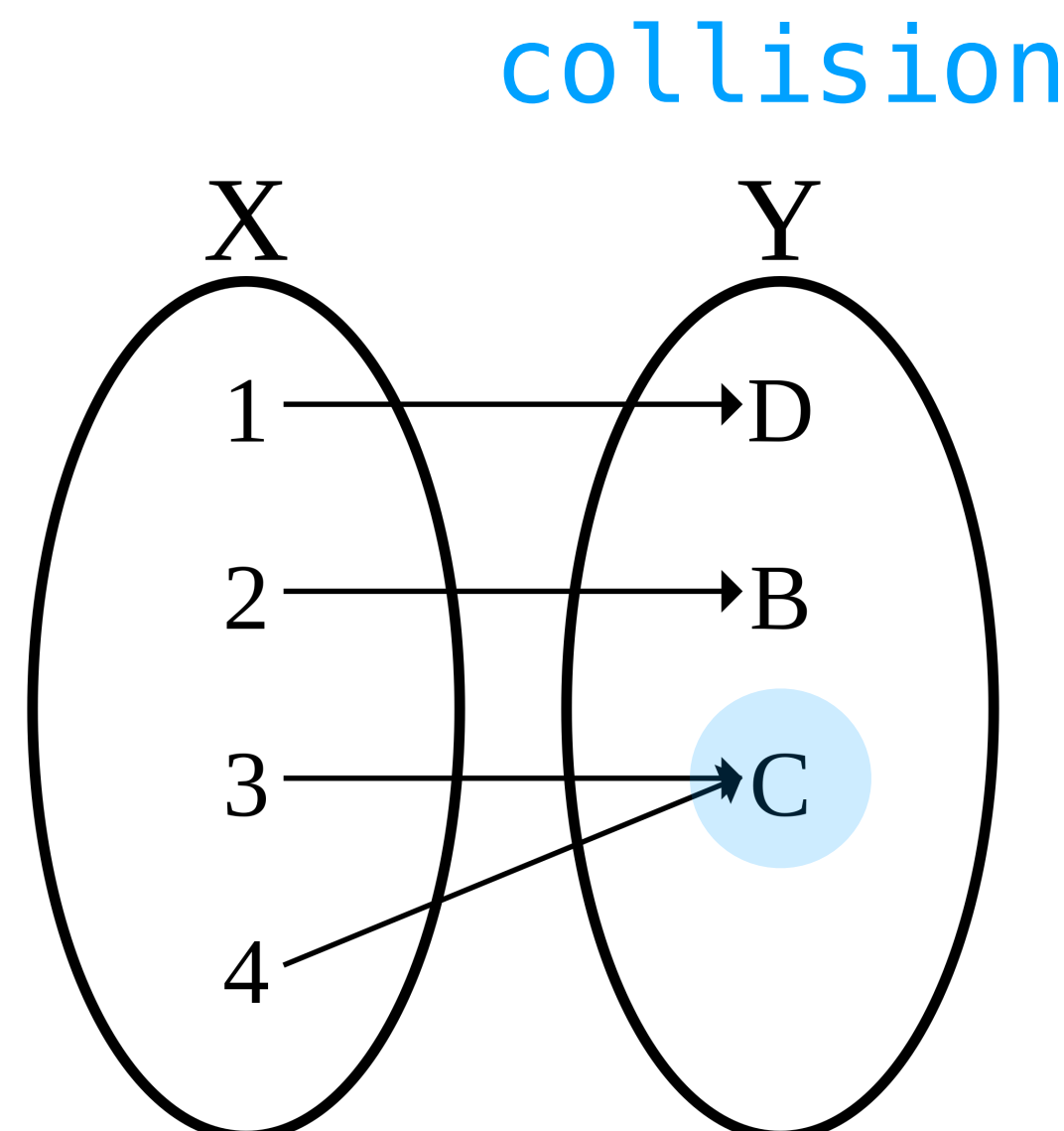
**Invertible transformations are 1-1 correspondences**



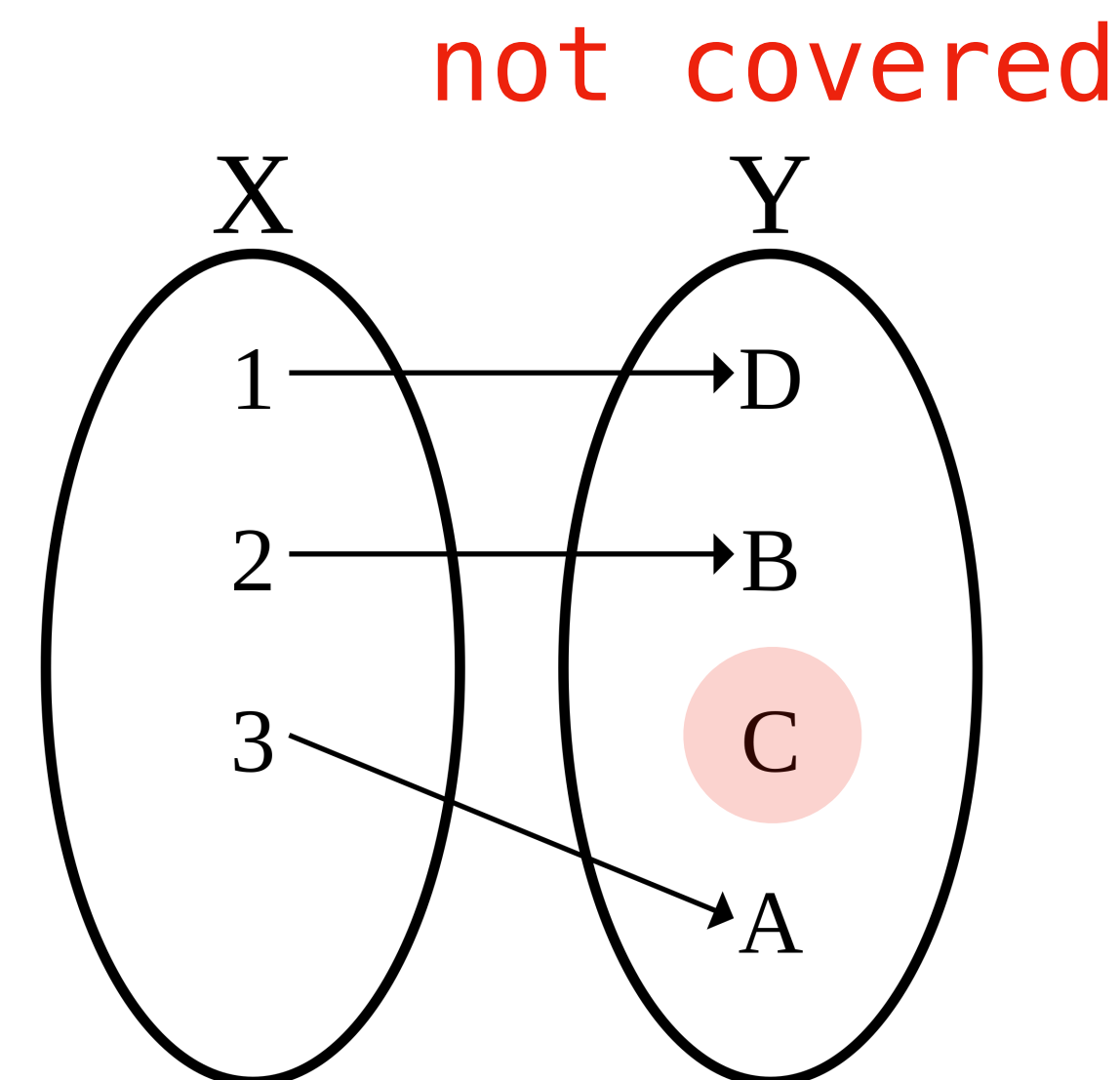
# Kinds of Transformations (Pictorially)



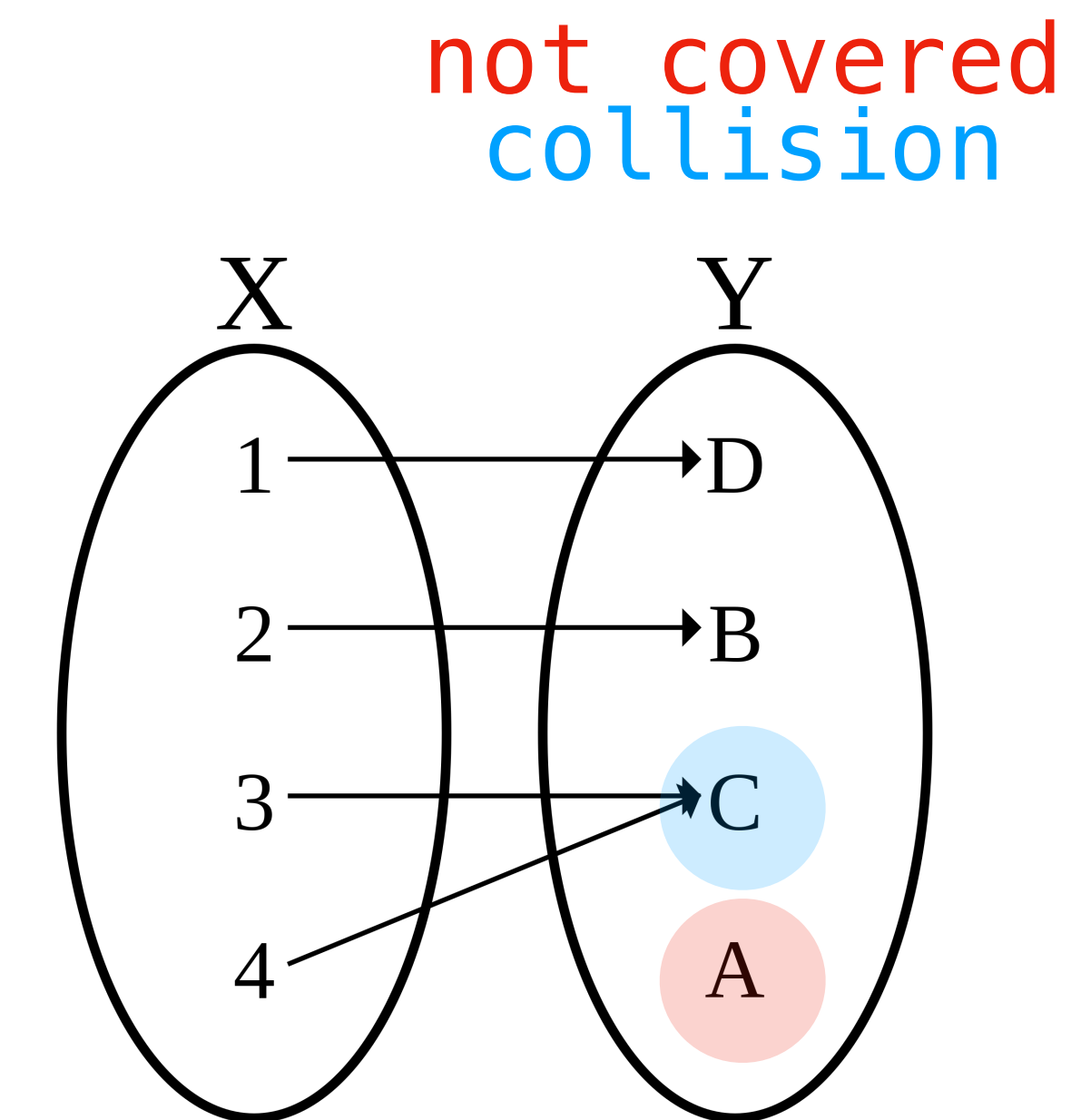
1-1 correspondence



onto, not 1-1



1-1 not onto



not 1-1, not onto

# Computing Matrix Inverses

# Fundamental Questions

How can we determine if a matrix has an inverse?

If a matrix has an inverse how do we compute it?



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Answer 1: Try to compute it

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Answer 2: Invertible Matrix Theorem (IMT)



**In General**

$$A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} = I$$

*Can we solve for each  $\mathbf{b}_i$ ?*

## In General

$$A\mathbf{b}_1 = \mathbf{e}_1$$

$$A\mathbf{b}_2 = \mathbf{e}_2$$

$$A\mathbf{b}_3 = \mathbf{e}_3$$

If we want a matrix  $B$  such that  $AB = I$ , then the above equation must hold (in the case  $B$  has 3 columns)

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If we want a matrix  $B$  such that  $AB = I$ , then the above equation must hold (in the case  $B$  has 3 columns)

**We need to solve 3 matrix equations**



# How To: Matrix Inverses

**Question.** Find the inverse of an invertible  $n \times n$  matrix  $A$

**Solution.** Solve the equation  $A\mathbf{x} = \mathbf{e}_i$  for every standard basis vector  $\mathbf{e}_i$ . Put those solutions  $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_n$  into a single matrix

$$[\mathbf{s}_1 \quad \mathbf{s}_2 \quad \dots \quad \mathbf{s}_n]$$

# How To: Matrix Inverses

**Question.** Find the inverse of an invertible  $n \times n$  matrix  $A$

**Solution.** Row reduce the matrix  $[A \ I]$  to a matrix  $[I \ B]$ . Then  $B$  is the inverse of  $A$

**This is really the same thing. It's a simultaneous reduction**

demo



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$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$



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The inverse is defined only if the determinant is nonzero

(see the notes on linear transformations for more information about determinants)



# Example

$$\begin{bmatrix} -6 & 14 \\ 3 & -7 \end{bmatrix}$$

# **Algebra of Matrix Inverses**

# Algebraic Properties (Matrix Inverses)

**Thm.** For an invertible matrix  $A$ , the matrix  $A^{-1}$  is invertible



# Algebraic Properties (Matrix Inverses)

**Theorem.** For an invertible matrix  $A$ , the matrix  $A^T$  is invertible

# Algebraic Properties (Matrix Inverses)

**Theorem.** For a invertible matrices  $A \in \mathbb{R}^{n \times n}$  and  $B \in \mathbb{R}^{n \times n}$ , the matrix  $AB$  is invertible



# Question

*Suppose that  $A$  is a  $n \times n$  invertible matrix such that  $A = A^T$  and  $B$  is a  $m \times n$  matrix*

*Simplify the expression  $A(BA^{-1})^T$  using the algebraic properties we've seen*

**Answer:**  $B^T$

$$A(BA^{-1})^T$$

$$A = A^T$$

# Invertible Matrix Theorem



# The Point

**Question.** How do we know if a square matrix is invertible?

**Answer.** *Every* perspective we've taken so far can help us answer this question

# Invertible Matrix Theorem

**Thm.** For any invertible matrix  $A \in \mathbb{R}^{n \times n}$

1.  $A^T$  is invertible

# Invertible Matrix Theorem

**Thm.** For any invertible matrix  $A \in \mathbb{R}^{n \times n}$

- 2.  $A\mathbf{x} = \mathbf{b}$  has at least one solution for every  $\mathbf{b}$
- 3.  $A\mathbf{x} = \mathbf{b}$  has at most one solution for every  $\mathbf{b}$
- 4.  $A\mathbf{x} = \mathbf{b}$  has at exactly one solution for every  $\mathbf{b}$



# Invertible Matrix Theorem

**Thm.** For any invertible matrix  $A \in \mathbb{R}^{n \times n}$

5.  $A$  has a pivot in every column

6.  $A$  has a pivot in every row

7.  $A$  is row equivalent to  $I_n$

# Invertible Matrix Theorem

**Thm.** For any invertible matrix  $A \in \mathbb{R}^{n \times n}$

- 8. The columns of  $A$  are linearly independent
- 9. The columns of  $A$  span  $\mathbb{R}^n$
- 10.  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution



# Invertible Matrix Theorem

**Thm.** For any invertible matrix  $A \in \mathbb{R}^{n \times n}$

- 11. The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is onto
- 12.  $\mathbf{x} \mapsto A\mathbf{x}$  is one-to-one
- 13.  $\mathbf{x} \mapsto A\mathbf{x}$  is a one-to-one correspondence
- 14.  $\mathbf{x} \mapsto A\mathbf{x}$  is invertible

# Taking Stock: IMT

*The following are logically equivalent:*

1.  $A$  is invertible
2.  $A^T$  is invertible
3.  $A\mathbf{x} = \mathbf{b}$  has at least one solution for any  $\mathbf{b}$
4.  $A\mathbf{x} = \mathbf{b}$  has at most one solution for any  $\mathbf{b}$
5.  $A\mathbf{x} = \mathbf{b}$  has a unique solution for any  $\mathbf{b}$
6.  $A$  has  $n$  pivots (per row and per column)
7.  $A$  is row equivalent to  $I$
8.  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution
9. The columns of  $A$  are linearly independent
10. The columns of  $A$  span  $\mathbb{R}^n$
11. The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is onto
12.  $\mathbf{x} \mapsto A\mathbf{x}$  is one-to-one
13.  $\mathbf{x} \mapsto A\mathbf{x}$  is a one-to-one correspondence
14.  $\mathbf{x} \mapsto A\mathbf{x}$  is invertible

These all express the  
same thing

!! only for square matrices !!

**We get a lot of information for free**



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**Thm.** If  $A$  is square, then

$A$  **is 1-1** if and only if  $A$  **is onto**

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**Warning.** Remember this only applies square matrices



# We get a lot of information for free

**Theorem.** If  $A$  is square, then

$A$  is invertible  $\equiv Ax = 0$  implies  $x = 0$

*Invertibility is completely determined by what  $A$  sends to  $0$*

# Question

*(True or False) If  $A$  is invertible, and  $B$  is row equivalent to  $A$  (we can transform  $B$  into  $A$  by a sequence of row operations), then  $B$  is also invertible*



# Question

*If  $[\mathbf{a}_1 \ \mathbf{a}_2 \ \mathbf{a}_3]$  is invertible, is  $[(\mathbf{a}_1 + \mathbf{a}_2 - 2\mathbf{a}_3) \ (\mathbf{a}_2 + 5\mathbf{a}_3) \ \mathbf{a}_3]$  also invertible?*

# Summary

The algebra of matrices can help us simplify matrix expressions

The invertible matrix theorem connects all the perspectives we've taken so far